

Project Demo Planning

Date	15 JULY 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine Using Machine Learning
Maximum Marks	1 Mark

Project Demo Planning:

A well-structured demo plan ensures that the team presents the project effectively, covering all key aspects in a clear and organized manner. This document outlines the plan for demonstrating the project, including the flow of the demo, key features to highlight, and responsibilities of each team member.

S.No	Demo Section	Description	Duration (mins)	Responsible Member
1	Problem Intro & Dataset Setup	Briefly introduce the crop recommendation problem and explain the Crop Recommendation dataset containing Nitrogen (N), Phosphorous (P), Potassium (K), temperature, humidity, pH level, rainfall, and crop labels.	2 mins	Hasini Badviti
2	Data Preprocessing & Cleaning	Demonstrate data cleaning, handling missing values, feature scaling, and preparing the agricultural dataset for machine learning.	3 min	Swarna Gera
3	Exploratory Data Analysis (EDA)	Present visualizations showing crop distribution, soil nutrient analysis, rainfall, temperature, humidity, and correlation heatmaps.	2 min	Manaswini Juluri
4	Machine Learning Model Selectio	Explain the preprocessing pipeline and compare the performance of KNN, Logistic Regression, Decision Tree, Random Forest, and K-Means. Describe how the best-performing model was selected.	3 min	Swarna Gera
5	Live Web Application Demonstration	Enter sample soil and environmental values into the web application and demonstrate real-time crop recommendation.	3 min	Hasini Badviti
6	Application Testing & Deployment	Demonstrate how the Flask application is executed locally, loads the trained machine learning model, and provides instant crop recommendations.	2 min	Manaswini Juuri

Demo Flow Summary:

Step	Activity	Notes
1	Introduction & Problem Statement	Explain the objective of recommending the most suitable crop using soil nutrients and environmental conditions to improve agricultural productivity..
2	Solution Overview	Summarize the system architecture, including data preprocessing, machine learning model, and Flask web app.
3	Live Feature Demonstration	Demonstrate the web application by entering sample soil parameters and displaying the recommended crop.

4	Q&A Session	Answer questions related to dataset preprocessing, ml model selection, prediction accuracy, and future improvements.
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